

Objective:

To demonstrate `IllegalArgumentException`

Execution Steps:

STEP 1:

Open IntelliJ IDEA.

STEP 2:

Create a Java project.

STEP 3:

Create a package named:

`javacore`

STEP 4:

Create a Java class named:

`IllegalArgumentException`

STEP 5:

Write the given program inside the class.

STEP 6:

Save the program.

STEP 7:

Compile the program.

STEP 8:

Run the program.

STEP 9:

Execution starts from the main() method.

STEP 10:

The try block starts executing.

STEP 11:

Thread.sleep(-100) is executed.

STEP 12:

The value -100 is an invalid argument because sleep time cannot be negative.

STEP 13:

IllegalArgumentException occurs.

STEP 14:

The catch block catches the exception.

STEP 15:

"Illegal Argument Exception" is displayed.

STEP 16:

"Program ended" is displayed.

STEP 17:

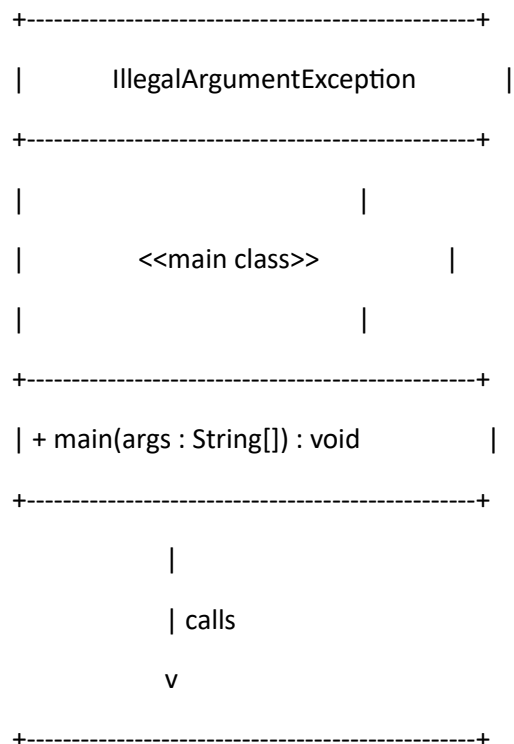
The program ends.

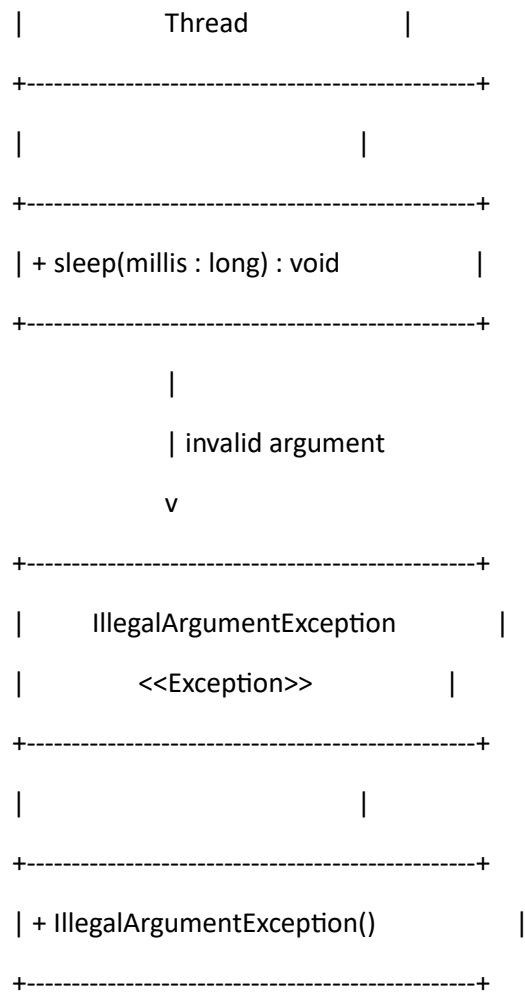
Algorithm:

1. Start the program.
2. Execute the main() method.

3. Start the try block.
4. Call Thread.sleep() with -100.
5. Check the argument passed to sleep().
6. Since -100 is an invalid argument,  
    IllegalArgumentException occurs.
7. Catch the exception using the catch block.
8. Display "Illegal Argument Exception".
9. Display "Program ended".
10. Stop the program.

UML Class Diagram:





Code:

```

package javacore;

public class IllegalArgumentException{
    public static void main(String[] args) {

        try {
            Thread.sleep(-100);
        }
        catch (IllegalArgumentException e) {
            System.out.println("Illegal Argument Exception");
        }
    }
}
  
```

```
        System.out.println("Program ended");  
    }  
}
```

Output:

Illegal Argument Exception

Program ended